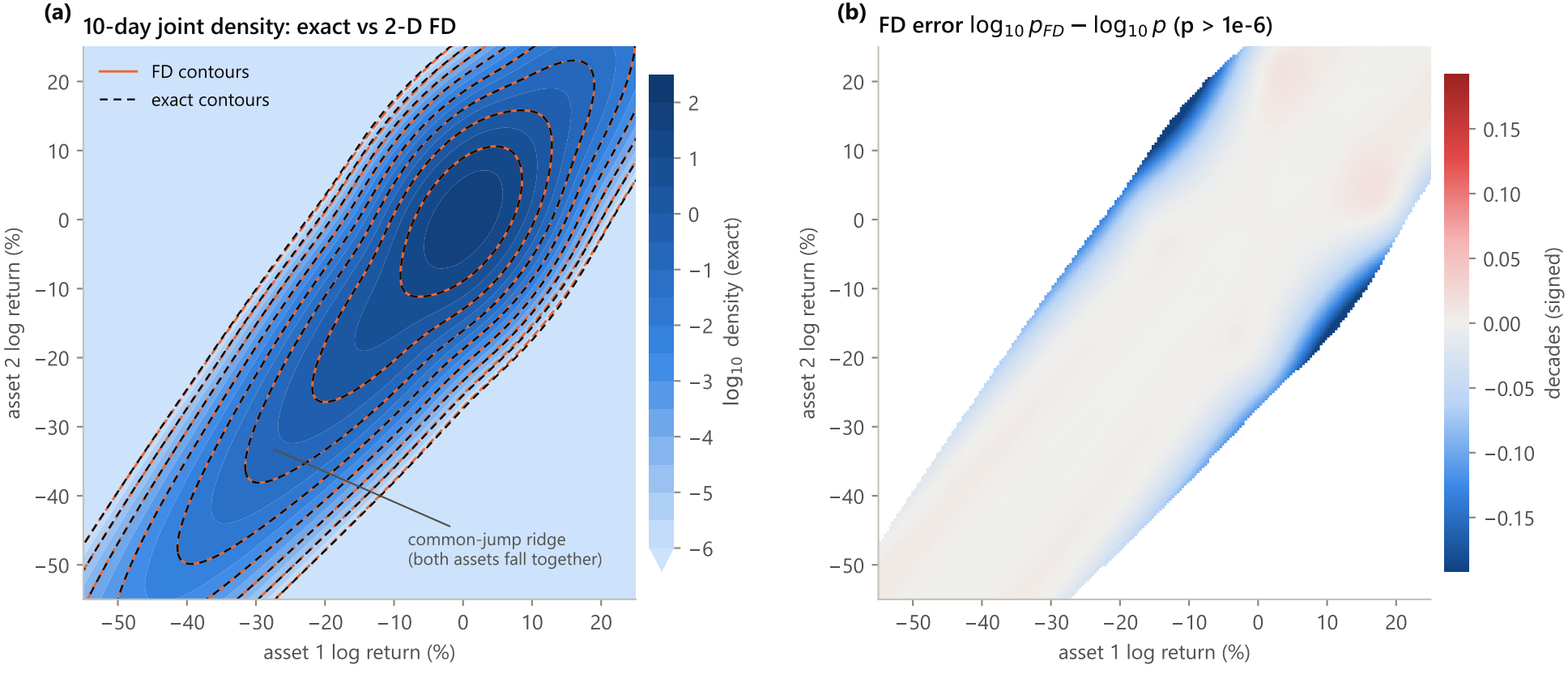

(a)
10-day joint density: exact vs 2-D FD
FD contours
exact contours
common-jump ridge
(both assets fall together)
asset 1 log return (%)
asset 2 log return (%)
log10 density (exact)
(b)
FD error log10 pFD − log10 p (p > 1e-6)
asset 1 log return (%)
asset 2 log return (%)
decades (signed)